

4. (a) Calcium reacts with fluorine to form calcium fluoride.

Showing the outer electrons only, draw dot and cross diagrams to show how bonding takes place during the formation of calcium fluoride.

Include the electronic structures and the charges of the ions formed.

[3]

calcium 2,8,8,2

fluorine 2,7

- (b) Potassium chloride and calcium oxide are both ionic compounds with high melting points.

The table gives information about the ions that make up each compound.

Ionic compound	Ions present
potassium chloride	potassium and chloride
calcium oxide	calcium and oxide

Calcium oxide has a higher melting point than potassium chloride.

Explain this difference.

[2]

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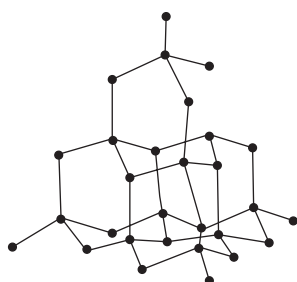
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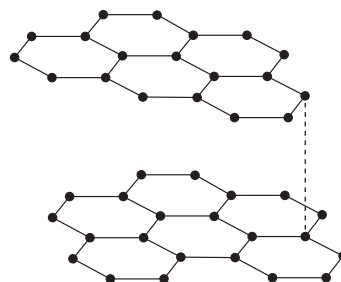
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(c) The diagrams show the structures of diamond and graphite.



diamond



graphite

Both structures have covalent bonds between carbon atoms.

Give **two** differences between the structures of diamond and graphite.

[2]

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